INCH-POUND
MIL-G-2857B
10 November 1988
SUPERSEDING
MIL-G-2857A
28 June 1968
(See 6.6)

MILITARY SPECIFICATION

GLASS, HEAT-TREATED, GLAZING, RECTANGULAR (FOR BRIDGE WINDOWS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers rectangular heat-treated quality clear glazing glass for bridge windows. This specification does not include bulletproof glass.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standard. The following specifications and standard form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-F-320 - Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

MILITARY

MIL-L-19140 - Lumber and Plywood, Fire-Retardant Treated.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5620
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STANDARD

MILITARY
MIL-STD-2073-1 - DoD Materiel Procedures for Development and Application of Packaging Requirements.

(Unless otherwise indicated, copies of federal and Military specifications and standards are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
C 1036 - Standard Specification for Flat Glass. (DoD adopted)
D 3951 - Standard Practice for Commercial Packaging. (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. The glass used for heat treatment shall be glazing quality, clear, polished plate glass (except as to nominal thickness and tolerances) in accordance with the requirements of ASTM C 1036, type I, class 1, quality q 3.

3.2 Construction. Heat treated glass shall be uniformly heated by any standard process capable of producing glass of the specified strength characteristics.

3.3 Tolerance. The rectilinear dimensions shall be as specified in the contract or order (see 6.2), with a tolerance of plus or minus 1/16 inch.

3.4 Finish. The corners shall be slightly rounded and all edges shall be ground or fire-polished to a neat finish.

3.5 Grooving. The glass shall be properly grooved or bevelled, as specified (see 6.2 and 6.3).

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3.6 Thickness. The thickness of the glass shall be 1/4, 5/16 or 3/8 inch, as specified (see 6.2). Individual finished sheets shall vary not more than 1/32 inch in thickness from the thickest to the thinnest part. The overall thickness shall not exceed 1/32 inch from the specified nominal thickness.

3.7 Impact resistance. Glass shall not crack or show any visible change (see 4.4.1).

3.8 Strain pattern. When viewed under polarized light, the glass shall show either concentric colored or black rings, or a uniformly distributed colored or black strain pattern, characteristic of fully heat-treated glass (see figure 1 and 4.4.3).

3.9 Light transmission. The heat-treated plate glass shall show a transmission of white light of not less than 86 percent (see 4.4.2).

3.10 Waviness. The glass shall exhibit no objectionable waves or similar objectionable defects as specified in ASTM C 1036 (see 4.4.4).

3.11 Workmanship. There shall be no defects in the finished glass beyond those allowed in the sheets before heat-treatment, except a slight warping such as may be experienced as a concomitance of the heat-treatment proper.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Quality conformance inspection. Quality conformance inspection shall consist of the examinations and tests specified in 4.3 and 4.4, respectively.

4.3 Visual and dimensional examinations. Samples shall be examined to determine compliance with 3.1, 3.2, 3.4, 3.10 and 3.11 for visual examination, and 3.3, 3.5 and 3.6 for dimensional examination.

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4.4 Tests. Samples shall be subjected to the tests specified in 4.4.1 through 4.4.4.

4.4.1 Impact test. Samples shall be inspected at the place of manufacture for conformance to 3.7. Material selected for impact test shall be placed on the frame described below and shall be subjected to the impact of a 1/2-pound steel ball striking a single gravity-accelerated blow at the center of the specimen from a height as shown in table I. The testing frame shall consist of an upper and lower piece of 1-inch thick iron plate, the inside dimensions of the rectangular frame shall be 12 by 12 inches; outside dimensions, 16 by 16 inches. The plates shall be cut from a 1-inch thick steel plate. The surface shall be smoothly planed and shall be covered with rubber gasket material 1/4 inch in thickness. The lower frame shall be bolted to a wooden frame of such thickness that the clearance between the under glass surface and the floor is 4 inches. Since heat-treated glass sheets cannot be cut to size, it is necessary to use the sheets as received. The lower frame shall be placed on the floor, and the sheet of glass shall be placed so that the "overhang" is nearly uniform. The upper frame shall then be placed with the rubber-protected edge down, so that the inner and outer edges coincide with the lower frame. The ball suspension shall be so placed as to drop the ball as near as possible to the center area of the glass plate bounded by the frame.

TABLE I. Impact test measurements.

Thickness of glass	Height of fall of ball
Inch	Feet
1/4	7
5/16	8-1/4
3/8	10

4.4.2 Light transmission test. Samples shall be tested at the place of manufacture. The rectangular (parallel) luminous transmittance of the sheet of glass shall be measured at normal incidence calculated to the International Commission on Illumination "Illuminant A". The original value shall be the photometer reading without the glass in position. The luminous transmittance after testing shall be considered as the percentage of the photometer readings with the glass in position as compared to the reading without the glass in position.

4.4.3 Polariscopic examination. Samples shall be subjected to polariscopic examination as specified in 3.8. When ordinary transparent material such as glass is subjected to stresses, the strained regions inside are characterized by changes in their optical properties, whereby they become partially birefringent. The amount of birefringence produced at any point in the material is, in part, a measure of the strain existing there. Consequently, when a sample glass containing strains is inserted between crossed polarizers, light is restored in

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the region under strain. This restored light usually appears in the form of various colored striations or colored patterns. If any glass does not show concentric rings as specified in 3.8, the lot represented shall be rejected. The strain pattern in properly heat-treated plate glass shall take the form of concentric rings of color near the edge of the plate of glass or may represent regular patterns of color (for example, Scotch plaid). The glass shall be moved to right or left as the pattern may not be visible at the particular angle in which the glass is placed with respect to the viewing polariscope used. A Nicol prism, or equal, or polaroid-equipped lens may be used, so long as the apparatus is equipped with a shutter mechanism carrying a tint plate so that the strain pattern may be viewed alternately as black lines or as a colored-fringed pattern. A typical black line pattern is illustrated on figure 1.

4.4.4 Waviness examination. Samples shall be visually examined under an equivalent of light from a clear sunless sky when placed in a vertical position plus or minus 10 degrees. Under such conditions, and with the viewer's eye on a level with the center of the sample, he shall look through the sample from a distance of 36 inches, toward a brick wall or similar background at the same distance, showing essentially straight parallel lines. During this examination, the viewer shall move his head horizontally through a slight distance to each side.

4.5 Inspection of packaging. Sample packs, and the inspection of the preservation, packing and marking for shipment, stowage, and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition.)

5.1 General.

5.1.1 Navy shipboard stowage fire-retardant requirements.

(a) Treated lumber and plywood. Unless otherwise specified (see 6.2), all lumber and plywood (including laminated veneer material used in shipping containers and pallet construction, members, blocking, bracing, and reinforcing) shall be fire-retardant treated material conforming to MIL-L-19140 as follows:

Levels A and B - Type II - weather resistant.
Category 1 - general use.
Level C - Type I - non-weather resistant.
Category 1 - general use.

(b) Fiberboard. Fiberboard used in the construction of class-domestic, non-weather resistant fiberboard, cleated fiberboard boxes (including interior packaging forms) shall meet the flame spread index and the specific optic density requirements specified in PPP-F-320.

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5.2 Preservation. Preservation shall be level A. Quantities of plate glass of uniform size lights, or a single light, shall be packaged in a snug-fitting fiberboard container conforming to PPP-B-636, class weather-resistant. Individual lights, and the top and bottom of the stack, shall be separated and overwrapped with a sheet of neutral unprinted paper. Containers shall be closed method V in accordance with the appendix to the box specification.

5.3 Packing. Packing shall be level A, B, C, or commercial as specified (see 6.2).

5.3.1 Levels A, B, and C containers. Glass preserved as specified in 5.2 shall be packed in overseas shipping containers in accordance with MIL-STD-2073-1, table VII of appendix C, as specified (see 6.2). The gross weight of a case of heat-treated glass shall not exceed 80 pounds.

5.3.2 Commercial. Glass preserved as specified (see 5.2) shall be packed for shipment in accordance with ASTM D 3951 and herein. The gross weight of a case of heat-treated glass shall not exceed 80 pounds.

5.4 Marking (levels A, B, C, and commercial). In addition to any special marking required (see 6.2), shipping containers shall be marked for shipment and storage in accordance with MIL-STD-2073-1, appendix F, and shall include bar codes and applicable packaging acquisition options herein as specified (see 6.2).

5.5 Special marking. The shipping container shall be plainly marked "HEAT-TREATED GLASS - HANDLE WITH CARE".

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The glass covered by this specification is intended to be used for bridge windows aboard ship and for general architectural use in construction.

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification.
(b) Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
(c) Rectilinear dimensions (see 3.3).
(d) Bevelling or grooving required (see 3.5).
(e) Thickness of glass desired (see 3.6).
(f) When fire-retardant treated lumber and plywood is not required (see 5.1.1(a)).
(g) Level of packing required (see 5.3).
(h) Additional packing requirements for level A, B, and C containers (see 5.3.1).
(i) Special marking, if required (see 5.4).
(j) Bar coding and packaging acquisition options (see 5.4).

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6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/ provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DoD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference paragraph	DID number	DID title	Suggested tailoring
3.5	DI-E-7031	Drawings, engineering and associated lists	----

The above DID was cleared as of the date of this specification. The current issue of DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.4 Suggested sampling for quality conformance inspection. Sampling for quality conformance inspection should be as specified in 6.4.1 through 6.4.3.

6.4.1 Lot. A lot should consist of all glass of one size produced in one plant under essentially the same conditions.

6.4.2 Sampling for visual and dimensional examinations. Samples for visual examination should be in accordance with level I of MIL-STD-105, and the acceptable quality level (AQL) should be 4.0 percent defective. Samples for dimensional examination should be in accordance with level S-3 of MIL-STD-105, and the AQL should be 2.5 percent defective.

6.4.3 Sampling for tests. Samples for quality conformance tests should be in accordance with level S-2 of MIL-STD-105. Failure of any test should be cause for rejection of the lot represented by the sample.

6.5 Subject term (key word) listing.

Grooving
Impact resistance
Light transmission
Polariscopic examination

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - ME
Navy - SH
Air Force - 84

Preparing activity:
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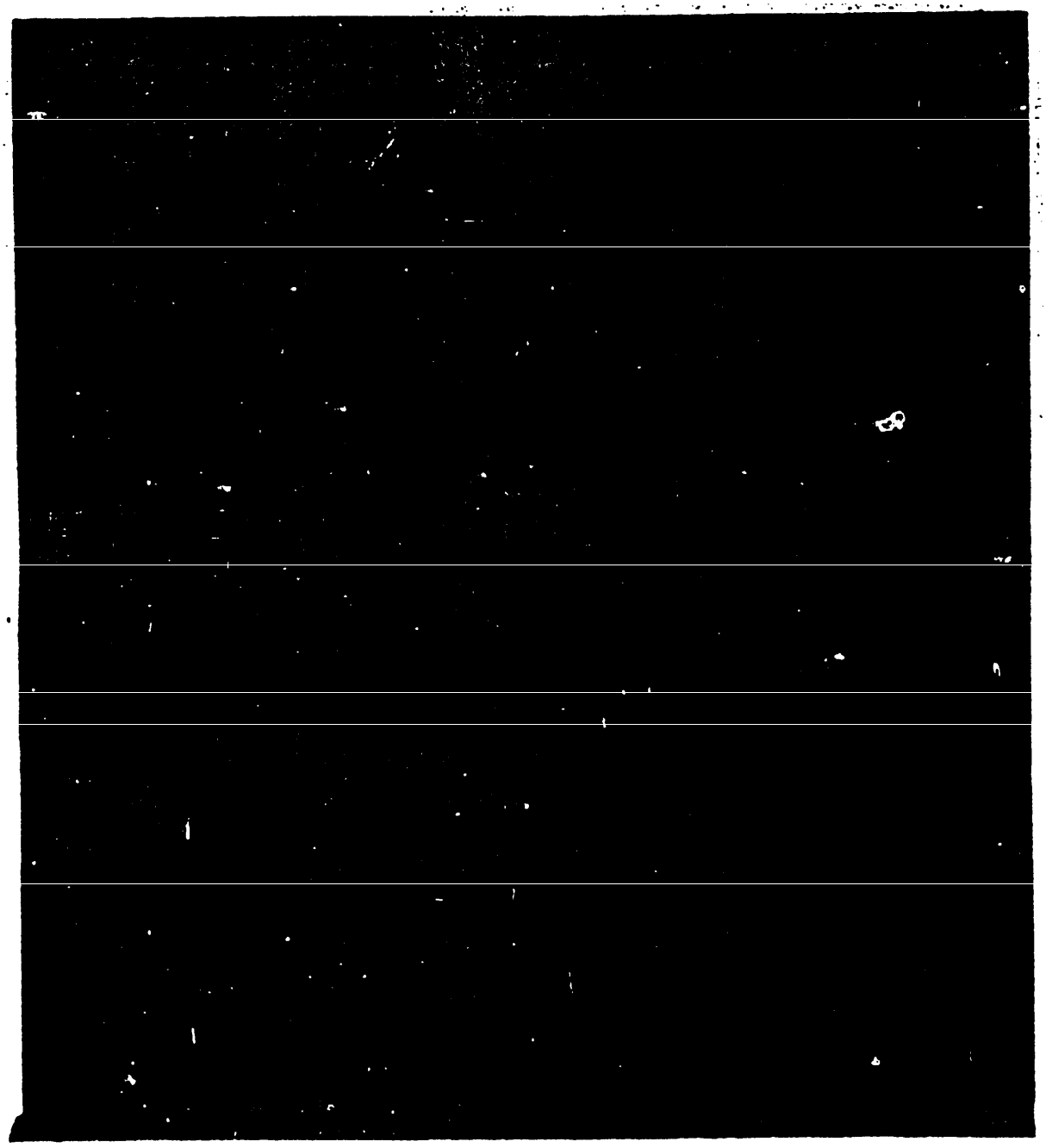


FIGURE 1. Strain pattern, 8-1/2 inches diameter, 7/8 inch thick.

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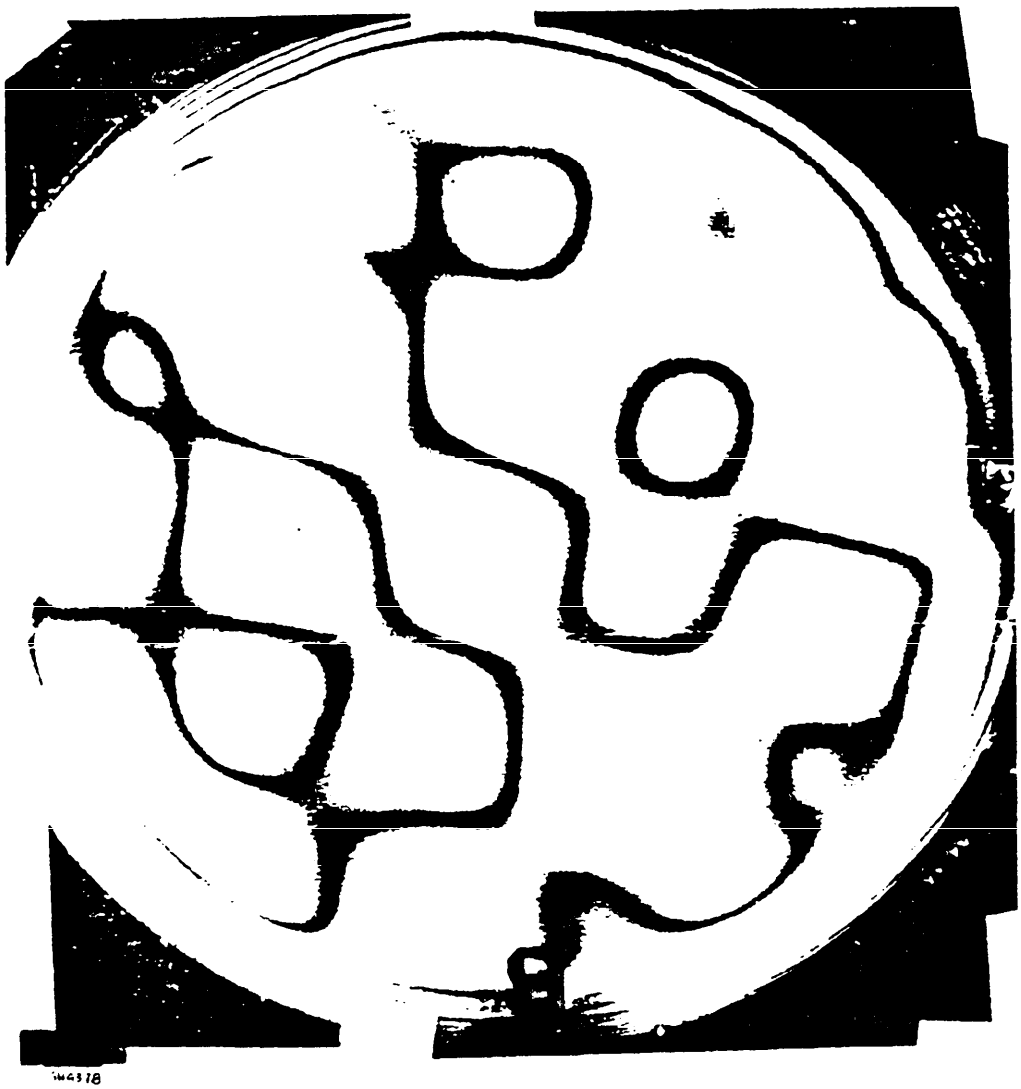


FIGURE 1. Strain pattern, 8-1/2 inches diameter, 7/8 inch thick - Continued

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions - Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-G-2857B	GLASS, HEAT-TREATED, GLAZING, RECTANGULAR (FOR BRIDGE WINDOWS)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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